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Selective Causal Adaptation for Evolving Time-Series Graphs: A Stability–Plasticity Study

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ABSTRACT Real-world multivariate time series rarely settle into a single stationary causal structure: physical, biological and financial systems undergo discrete regime changes during which only a subset of their causal relationships shifts while the remainder persists. Existing dynamic causal-discovery frameworks (CD-NOD, RPCMCI, CASTOR, FANTOM) adapt at the level of the entire graph, re-estimating a full structure only once a regime boundary is detected. This paper introduces selective causal adaptation (SCA), a lightweight, training-free streaming estimator which instead adapts each candidate lagged edge independently. SCA maintains a persistent causal-graph memory state and updates it with an edge-specific plasticity rate computed as the product of a global change-intensity signal, a saturating local-deviation signal and an edge-specific persistence score, ensuring aggressive updates only occur where a detected structural change coincides with a sustained local disagreement. We evaluate SCA against a memory-free Rolling-OLS baseline and a regime-aware CD-NOD baseline on a recurring-regime synthetic benchmark ($T = 5000$, three change points, $d \in \{4, 10, 20, 50\}$, $L \in \{1, 2, 4\}$, four noise scales and three noise distributions), a development ablation and an external Causal Chambers benchmark. Under the canonical dynamic condition and across the factorial design, SCA improves precision and F1 against the Rolling OLS (pooled F1 difference $+0.0498$, Holm-corrected $p < 0.001$) at the cost of a small decrease in recall while CD-NOD achieves a higher F1 than SCA in the canonical condition (0.775 vs. 0.637); the overall adaptation-delay difference between SCA and Rolling OLS is not statistically significant ($p = 0.712$). Sensitivity analysis and a high-dimensional, high-lag stress test ($d = 50$, $L = 4$) additionally reveal a detector failure mode in which change detection collapses to 0% despite unchanged false-alarm behaviour elsewhere. We report these results, and where SCA does not dominate, discuss the resulting stability–plasticity trade-off, the method’s scope as a Granger-style, past-only estimator and the extensions required for wider application.

INDEX TERMS Causal discovery, dynamic causal graphs, edge-level plasticity, Granger causality, non-stationary time series, regime-change detection, stability–plasticity trade-off, structural Hamming distance.

I. INTRODUCTION

MULTIVARIATE time series generated by physical, biological, and financial systems rarely remain governed by a single, fixed causal structure. Cortical connectivity re-organises during a seizure, a wind-tunnel chamber’s causal structure shifts once an actuator intervenes, and macroeconomic relationships change at business-cycle turning points. In each case the underlying causal graph does not merely add noise around a fixed structure; it changes at discrete, a priori unknown points in time, and the change is frequently localised: a handful of edges are perturbed while the remainder of the mechanism is untouched.

Classical causal-discovery methods such as PCMCI+ and

DYNOTEARS assume a single stationary regime and degrade once the structure drifts. A second family of methods addresses non-stationarity directly by discovering regimes: CD-NOD detects change points and returns a single summary graph; RPCMCI learns a graph per regime but requires the regime count fixed in advance and is restricted to lagged edges; CASTOR and FANTOM jointly infer a regime partition and a separate directed graph per regime, with FANTOM additionally handling heteroscedastic and non-Gaussian noise through conditional normalising flows within a Bayesian expectation-maximisation loop. All of these regime-aware approaches formulate adaptation at the level of the graph as a whole: once a boundary is declared, a

new structure is estimated, implicitly discarding the previous topology. This is a reasonable strategy when a regime shift touches most of the mechanism, but it is a poor match for the common case of localised non-stationarity, where re-estimating every edge (i) discards historical knowledge that remains valid, (ii) introduces adaptation latency because a full topology needs enough fresh samples to be re-estimated reliably, and (iii) makes the system sensitive to a single noisy window being mistaken for a genuine change.

This paper studies an alternative unit of adaptation: the individual edge rather than the whole graph. We introduce Selective Causal Adaptation (SCA), a streaming, training-free estimator that maintains a persistent causal-graph memory and updates each candidate lagged edge with its own plasticity rate. The plasticity rate for an edge is high only when three conditions hold simultaneously: a structural change has been detected somewhere in the graph, the edge's freshly estimated value disagrees with its remembered value, and that disagreement has persisted rather than appearing in a single noisy window. When any of these signals is absent, the edge's rate collapses toward a small floor and its memory entry is left essentially undisturbed. This design directly targets the stability–plasticity trade-off central to online causal discovery: memory should smooth stable relationships but react promptly to changed ones.

SCA is deliberately implemented on top of a transparent, regression-based, past-only (Granger-style) evidence score rather than a trained neural encoder. This keeps the pipeline free of gradient-based training, makes every design decision traceable to a closed-form update rule, and keeps the computational footprint small enough for streaming or embedded causal monitoring.

The paper asks three research questions. **RQ1:** Does persistence-gated memory improve the stability of causal graph estimates under recurring regime changes, relative to a memory-free rolling estimator? **RQ2:** Does edge-level selective plasticity improve adaptation relative to a global (graph-wide) update rate? **RQ3:** How does SCA compare with a memory-free baseline (Rolling OLS) and a regime-aware baseline (CD-NOD) across a broad factorial design spanning dimensionality, lag order, noise scale, and noise distribution, and where does it fail?

We evaluate these questions on a controlled recurring-regime synthetic benchmark, a four-variable development/ablation benchmark, and an external Causal Chambers physical benchmark. Consistent with the goal of an honest evaluation, we report results in both directions: SCA improves precision, pooled F1, and (in the development ablation) stable-edge preservation and adaptation delay relative to Rolling OLS, but CD-NOD attains a higher F1 in the canonical dynamic condition, the aggregate adaptation-delay improvement over Rolling OLS is not statistically significant, and a high-dimensional, high-lag stress test exposes a change-detector failure mode. We treat these as central findings rather than as caveats to be minimised.

Contributions. This paper makes the following contributions:

- (1) a graph-memory framework for evolving lagged causal graphs that separates a local, past-only evidence estimator from a persistent memory tensor;
- (2) a persistence-gated, edge-level plasticity mechanism combining a global change-intensity signal, a saturating local-deviation signal, and an edge-specific persistence score into a single multiplicative update rate;
- (3) a controlled factorial evaluation across dimension ($d \in \{4, 10, 20, 50\}$), lag order ($L \in \{1, 2, 4\}$), noise scale, and noise distribution, together with a paired statistical-testing protocol;
- and (4) a detailed, transparent analysis of the stability–plasticity trade-offs and failure modes the mechanism exhibits, including a high-dimensional detector-failure regime. We phrase these contributions to match what the implementation and experiments actually support, and we do not claim that SCA outperforms all baselines or achieves state-of-the-art dynamic causal discovery.

II. RELATED WORK

Causal discovery from stationary time series is a mature area. PCMCI and its extension PCMCI+ [1] combine a condition-selection stage with conditional-independence testing to recover lagged and (in PCMCI+) contemporaneous links, scaling to moderately high dimensions but assuming a single stationary regime. DYNOTEARS [2], [3] casts structure learning as continuous, acyclicity-constrained optimisation over both instantaneous and lagged adjacency matrices, building on the NOTEARS acyclicity characterisation. Rhino [4] relaxes the constant-variance assumption to history-dependent noise but still fits a single graph to the whole series. All three approaches degrade once the generating mechanism itself changes over time, motivating the regime-aware literature discussed next.

A second line of work targets non-stationarity directly by discovering regimes. CD-NOD [5] treats the time index as a surrogate variable, detects change points through the variability of conditional distributions, and returns a single summary graph whose parents are the union of per-regime parents; it does not recover individual per-regime graphs or the number of regimes, and it assumes homoscedastic noise. RPCMCI [6] learns a temporal graph per regime but requires the number of regimes and transition points to be specified in advance and restricts attention to lagged relationships. CASTOR [7] jointly infers regime indices and a directed graph per regime, including both instantaneous and lagged edges, but assumes Gaussian noise with equal variance across regimes (equivariance). FANTOM [8] generalises this line further with a Bayesian expectation-maximisation procedure over conditional normalising flows, jointly inferring the number of regimes, their indices, and per-regime DAGs under either non-Gaussian or heteroscedastic noise, and reports consistent gains over CASTOR, RPCMCI, CD-NOD, and stationary baselines on synthetic and real data (wind-tunnel and EEG). SPACETIME [9] similarly groups time periods sharing an invariant mechanism. Separately, temporal graph attention networks [10] show that dynamic graph structure can be encoded efficiently from streaming interactions,

though they target link prediction rather than causal discovery; an early internal iteration of the present project explored a similar deep temporal-graph-attention encoder combined with conditional-spline flows, mirroring FANTOM's machinery, before we deliberately replaced it with the transparent regression-based estimator described in Section IV.

The common feature of CD-NOD, RPCMCI, CASTOR, and FANTOM is that adaptation is formulated at the level of the whole graph (or, for CD-NOD, a single summary graph): a regime boundary triggers estimation of a new structure. This is well matched to settings where a regime shift touches most of the mechanism, and FANTOM in particular provides identifiability guarantees that the present, purely empirical study does not attempt to match. SCA instead targets the complementary setting of localised non-stationarity, in which only a subset of edges is perturbed at a change point, by moving the unit of adaptation to the individual edge and combining a lightweight two-window change detector with an edge-specific persistence tracker. This design trades the richer generative assumptions and identifiability results of FANTOM-style models for a training-free, closed-form update rule with a small, fixed memory footprint.

Benchmarking resources for this literature include the TimeGraph synthetic suite [11], the physical Causal Chambers wind-tunnel testbed [12], and EEG seizure corpora documented to exhibit heteroscedastic sensor noise [8], [13]; we draw on the Causal Chambers wind-tunnel data for the external benchmark reported in Section VI-F. Identifiability of additive-noise causal models under the causal Markov, minimality, sufficiency, and (within-regime) stationarity assumptions is established in [14] and extended to heteroscedastic and multi-regime settings in [8]; SCA relies on this line of results only informally, as a justification for using a regression-based evidence score as a proxy for edge presence, rather than proving a new identifiability result of its own.

III. PROBLEM FORMULATION

Let $X = \{x_t\}_{t=1}^T$ denote a d -dimensional multivariate time series, $x_t \in \mathbb{R}^d$. A time-varying causal graph is represented by a tensor $G_t \in \mathbb{R}^{d \times d \times (L+1)}$, where the third index is the lag $\ell \in \{0, 1, \dots, L\}$. We assume a piecewise-constant generating graph with unknown regime-change points τ , so that between consecutive change points the graph is fixed and equal to some regime graph (denoted A for the regime immediately before a transition and B for the regime immediately after). We define the **stable edge set** and the **changed edge set** at a transition as

$$E_{\text{stable}} = B \cap A, \quad (1)$$

$$E_{\Delta} = (B \setminus A) \cup (A \setminus B), \quad (2)$$

so that E_{Δ} collects every edge that is added or removed by the transition, and E_{stable} collects every edge whose lagged coefficient is unchanged. The estimation problem is to produce, at every time step t , an estimate that jointly satisfies two objectives:

Stability: for $(i, j, \ell) \in E_{\text{stable}}$, the estimate at time t should continue to match its pre-transition value once the transition has occurred, i.e., the memory should not be disturbed by a change elsewhere in the graph.

Plasticity: for $(i, j, \ell) \in E_{\Delta}$, the estimate should converge to its post-transition value as quickly as possible once the transition has occurred.

These two objectives are in direct tension: a memory mechanism aggressive enough to track E_{Δ} quickly will, absent an edge-specific gating signal, also perturb E_{stable} , while a memory mechanism conservative enough to fully protect E_{stable} will adapt E_{Δ} slowly. Section IV formalises SCA as a mechanism that resolves this tension at the level of the individual edge (i, j, ℓ) rather than through a single global learning rate. We also note explicitly, consistent with the discussion in Section VIII, that the estimator considered here is a past-only (Granger-style) predictive estimator: it regresses each target on lagged predictors only, the lag-0 (contemporaneous) plane is fixed to zero throughout, and no claim is made that the identified relationships correspond to contemporaneous structural causal effects in the sense of Pearl's do-calculus.

IV. SELECTIVE CAUSAL ADAPTATION

SCA processes the series causally: only samples up to and including the current time step are used at every stage. The pipeline (Fig. 1) has six components run once per time step over a trailing window of length W : a local graph estimator, a two-window change detector, an edge-level deviation and persistence tracker, a plasticity gate, and a memory update. Fig. 2 illustrates how the resulting plasticity rate separates the fate of a stable edge from that of a changed edge.

A. LOCAL GRAPH ESTIMATION

Within a trailing window, the variables are standardised and a joint multivariate least-squares (OLS) model is fit:

$$x'_t = \beta_0 + \sum_{\ell=1}^L B_{\ell} x'_{t-\ell}, \quad (3)$$

and the local graph estimate is read off as the regression coefficients,

$$\hat{G}_{ij\ell,t} = [B_{\ell}]_{ij} \quad (4)$$

for lags $\ell = 1, \dots, L$. The lag-zero plane is fixed to zero throughout, since conditioning on $X_i(t)$ to predict $X_j(t)$ is not identifiable from observational data without further structural assumptions and would additionally introduce contemporaneous-leakage artefacts in a streaming setting; diagonal (self-loop) entries are likewise removed. Consequently $\hat{G}$ is a past-only, Granger-style estimator: it reflects predictive, lagged dependence, not necessarily an instantaneous structural mechanism.

B. CHANGE DETECTION

To decide whether the series is currently undergoing a genuine, graph-wide structural change, two adjacent, non-overlapping windows of length D immediately preceding t are

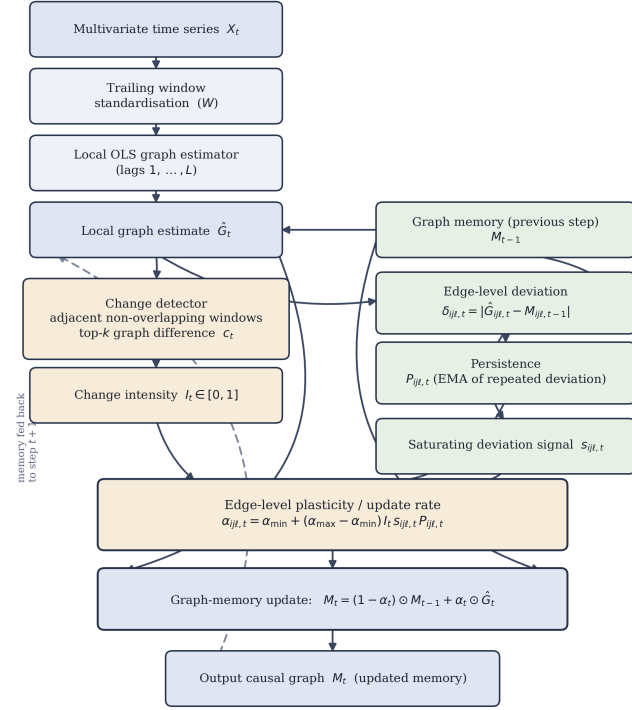


FIGURE 1. Overall SCA architecture. A local OLS graph estimate is compared against graph memory; a two-window change detector and an edge-level deviation/persistence pathway jointly gate a per-edge plasticity rate that updates the memory at every step.

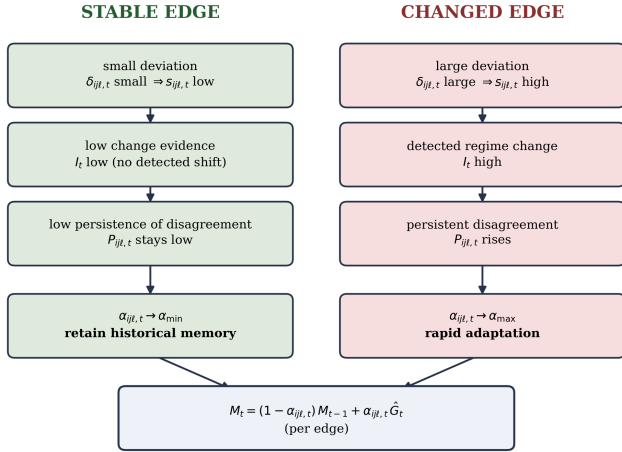


FIGURE 2. Stability-plasticity mechanism. A stable edge accumulates low deviation, change intensity, and persistence, so its plasticity collapses toward $\alpha_{\min}$; a changed edge accumulates high values of all three and its plasticity approaches $\alpha_{\max}$.

each summarised into a local graph, and the k largest entries of their absolute difference (after zeroing the diagonal) are averaged into a scalar change score:

$$c_t = \text{mean}(\text{top}_k(|\hat{G}_{\text{rec},t} - \hat{G}_{\text{prev},t}|)) \quad (5)$$

with $k = \min(4, d^2(L+1))$. Using only the strongest few edge deviations, rather than an average over the entire graph, keeps the statistic sensitive to a small number of genuinely perturbed edges without being diluted by the $d^2(L+1)$ entries

that typically remain unchanged. The corresponding global change intensity is

$$I_t = \begin{cases} \min(1, c_t/\theta) & \text{if } c_t \geq \theta, \\ 0 & \text{otherwise,} \end{cases} \quad (6)$$

where θ controls detector sensitivity and $I_t \in [0, 1]$ represents the strength of any detected structural change. $I_t = 0$ whenever no change is flagged, so that no edge receives a boosted plasticity rate in the absence of a detected shift.

C. EDGE-LEVEL DEVIATION AND PERSISTENCE

For every candidate edge (i, j, ℓ) , the deviation between the freshly estimated local graph and the current memory is

$$\delta_{ij\ell,t} = |\hat{G}_{ij\ell,t} - M_{ij\ell,t-1}|. \quad (7)$$

Persistence captures repeated evidence that an edge differs from its remembered value, via an exponential moving average of an indicator that the deviation exceeds a small tolerance ϵ :

$$P_{ij\ell,t} = \lambda P_{ij\ell,t-1} + (1 - \lambda) \mathbb{I}[\delta_{ij\ell,t} \geq \epsilon], \quad (8)$$

so a single noisy window cannot, by itself, push an edge's persistence score to a high value; P rises only when the disagreement between the local estimate and memory is sustained across several consecutive steps. Finally, the raw deviation is mapped to a bounded, saturating signal,

$$s_{ij\ell,t} = \frac{\delta_{ij\ell,t}}{\delta_{ij\ell,t} + \epsilon} \in [0, 1), \quad (9)$$

which grows quickly for small deviations and saturates for large ones, preventing an occasional very large outlier from dominating the plasticity rate.

D. PLASTICITY UPDATE

The three signals above—whether the graph as a whole is changing (I_t), whether this particular edge is currently deviating from memory ($s_{ij\ell,t}$), and whether that deviation has persisted ($P_{ij\ell,t}$)—are combined multiplicatively into a single edge-level plasticity (update) rate, mapped into an admissible range $[\alpha_{\min}, \alpha_{\max}]$:

$$\alpha_{ij\ell,t} = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) I_t s_{ij\ell,t} P_{ij\ell,t}, \quad (10)$$

with diagonal entries then set to zero. This multiplicative structure is central to the mechanism: aggressive updating of an edge occurs only when (1) a structural change has been detected somewhere in the graph, (2) the local estimate disagrees with graph memory for this specific edge, and (3) that disagreement has persisted across recent steps. If any one of the three factors is near zero, the product collapses toward $\alpha_{\min}$ and the edge's memory entry is barely disturbed, which is precisely the mechanism that protects an unrelated stable edge from a regime change occurring elsewhere in the graph. Conversely, $\alpha_{\min} > 0$ is set strictly positive so that even edges not flagged by the detector retain a small, non-zero rate of slow adaptation (drift), rather than being frozen indefinitely.

E. GRAPH-MEMORY UPDATE

The memory tensor is updated using a per-edge exponential moving average, controlled by the edge-level plasticity rate:

$$M_t = (1 - \alpha_t) \odot M_{t-1} + \alpha_t \odot \hat{G}_t, \quad (11)$$

where $\odot$ denotes element-wise multiplication and the memory tensor M_t is emitted directly as the model's causal-graph prediction at time t . A high α leads to fast changes and weak memory retention; a low α leads to strong retention of the past estimate. Since α is calculated separately for each edge, stable edges retain their information even when they lie alongside, or coincide in time with, edges that are changing quickly because of a real change.

F. COMPUTATIONAL CONSIDERATIONS

SCA never trains a neural network, so its per-step state is limited to the memory and persistence tensors, each of size $O(d^2(L+1))$. Its dominant per-step cost is $d(d-1)(L+1)$ small OLS fits over a window of W samples for the local estimator, plus an $O(d^2(L+1))$ comparison for the two-window change detector, both of which scale polynomially rather than requiring iterative optimisation or gradient-based training. Runtime therefore grows with window size W (larger windows cost more per fit but stabilise the local estimate), with the number of variables d (quadratically, through the number of candidate edges), with the lag order L (linearly in the number of lag planes, but see Section VI-G for the disproportionate effect of L on the change detector), and with the update frequency U at which the memory is refreshed. We do not report a closed-form asymptotic complexity beyond this qualitative scaling, since the constant factors are dominated by the underlying linear-algebra implementation rather than by the SCA update rule itself.

V. EXPERIMENTAL METHODOLOGY

A. SYNTHETIC DATA

A recurring-regime vector autoregressive (VAR) generator is employed in the primary evaluation. Each of the generated time series consists of $T = 5000$ observations with alternating regimes $A \rightarrow B \rightarrow A \rightarrow B$. The change points are set at times $t = 1000, 2500$, and 4000 (Fig. 3). The overall setup comprises connecting each output variable to a lagged variable with a coefficient of 0.30. With each regime transition, one previous lagged connection is withdrawn, and a new lagged connection is established. Innovations are scaled down by 0.1 prior to the application of the noise scale factor outlined below. The repetition of regime A in the interval $t \in [2500, 4000]$ is intentional since it enables us to determine whether the presence of memory affects the speed of adaptation to the recurring environment.

The factorial design combines noise scale $\sigma \in \{0.05, 0.10, 0.25, 0.50\}$, noise distribution $\in \{\text{Gaussian, unit-variance Student-}t, \text{uniform}\}$, dimensionality $d \in \{4, 10, 20, 50\}$, and maximum lag order $L \in \{1, 2, 4\}$. This results in a total of 144 factorial cells ($4 \times 3 \times 4 \times 3$;

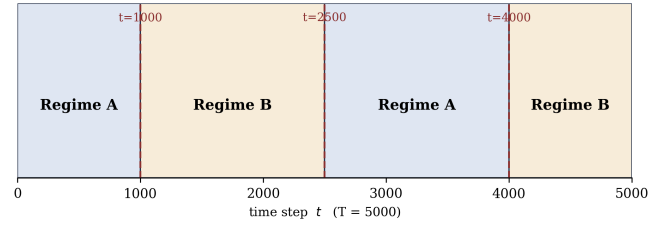


FIGURE 3. Recurring-regime timeline used by the synthetic generator. The return to Regime A at $t = 2500$ – 4000 tests whether persistent memory accelerates recovery of a previously seen structure.

equivalently, 288 method-by-condition cells when SCA and Rolling OLS results are considered together). Hyperparameter selection and sensitivity analysis (Section VI-C) use seeds 0–19; all main results from Section VI are based on the held-out seeds 20–39 (i.e., 20 seeds) unused for tuning decisions. The chronological order is maintained, so there is no test-set leakage: no sample from the future is used to produce a prediction at an earlier time.

B. BASELINES

The canonical dynamic comparison is SCA versus Rolling OLS versus CD-NOD. Rolling OLS re-estimates the graph at every step using the same trailing-window OLS procedure as SCA's local estimator. It includes no change detection, maintains no persistence, and has no memory. Its output at time t is simply $\hat{G}_t$ thresholded using the same edge-decision rule as SCA. This setup isolates the contribution of the memory-and-plasticity mechanism by keeping the estimator fixed. CD-NOD is evaluated directly on the pre-specified canonical dynamic cell (Section VI-A) using Fisher-Z conditional-independence tests and a chronological domain index. A full CD-NOD sweep across all 144 factorial cells was not computationally feasible and was not attempted. PCMCi+ is evaluated as a reference using the external Causal Chambers benchmark (Section V-F). RPCMCi is implemented where appropriate, as an adapter. CASTOR and FANTOM are mentioned as related regime-aware approaches (Section II), but they were not used in this study's factorial design, and no fabricated results are provided for methods that were not actually run.

C. EVALUATION METRICS

The main goal is to measure F1 score based on whether an edge exists or not. An edge is considered present if its estimated weight goes above a fixed threshold, which we set beforehand as θ_{edge} . We also look at other measures. Precision tells us how many of the edges we say are present actually are present. Recall shows how many of the true edges we manage to find. Structural Hamming distance (SHD) counts up all the mistakes—edges that are wrongly added or missed entirely. Adaptation delay is when we first see all the changes correctly identified after a shift happens. Stable-edge preservation checks how often, after the change, every edge that should stay the same is still predicted correctly.

Change-detection rate measures how often we correctly spot a change after it occurs. False-alarm rate tracks how often we wrongly think a change happened during stable periods or in experiments where no real change took place.

D. STATISTICAL TESTING

Paired comparisons between SCA and Rolling OLS are conducted with two-sided paired sign-flip randomisation tests (100,000 draws), reported alongside paired bootstrap 95% confidence intervals for the mean difference and paired Cohen's d_z effect sizes. All reported comparisons across metrics are Holm-corrected for multiple testing [15]. SCA is compared against CD-NOD only on the pre-specified canonical dynamic cell, without a paired statistical test across a factorial sweep, since CD-NOD was not run on the full design.

E. HYPERPARAMETER SELECTION

SCA's hyperparameters were set to fixed values before the held-out evaluation was performed. The values used were: change-detector window $D = 50$, local-estimation window $W = 200$, detection threshold $\theta = 0.30$, persistence decay $\lambda = 0.97$, edge-change tolerance $\epsilon = 0.10$, plasticity bounds $[\alpha_{\min}, \alpha_{\max}] = [0.02, 0.50]$, memory update frequency $U = 10$, and edge-decision threshold $\theta_{\text{edge}} = 0.15$. The effect of each of these hyperparameters on F1, SHD and false-alarm rate is studied in the one-factor-at-a-time sensitivity analysis in Section VI-C.

F. EXTERNAL BENCHMARK

In addition to the factorial design, SCA and PCMCi+ are tested on the `wt_walks_v1` / `actuators_random_walk_2` series from the Causal Chambers wind-tunnel testbed [12]. This series has 32 variables that are aligned with a graph structure and contains 10,000 observations arranged in order. The lags from 1 to 10 are combined into a single lag-agnostic task for directed-edge recovery. This is not a benchmark suite but a single physical data series. The results, found in Section VI-F, are presented as an illustrative validation rather than as proof of reliability across real-world situations.

VI. RESULTS

A. CANONICAL DYNAMIC COMPARISON

Table 1 presents the results for the pre-specified canonical dynamic cell ($d = 4$, $L = 2$, Gaussian noise, $\sigma = 0.10$) comparing three methods: SCA, Rolling OLS and CD-NOD. SCA achieves higher precision than both baseline methods and shows a shorter adaptation delay compared to CD-NOD. However, CD-NOD reaches the highest F1 score among the three at 0.775, while SCA scores 0.637 and Rolling OLS scores 0.617. This better F1 performance comes from CD-NOD's higher recall (1.000 versus 0.551 for SCA). In addition, CD-NOD delivers perfect stable-edge preservation and flawless change detection in this cell. SCA, on the other hand, has a detection rate of 1.000 ± 0.000 with a false-alarm rate of 0.091 ± 0.001 . We state these findings directly rather than framing precision or delay as an overall win. On

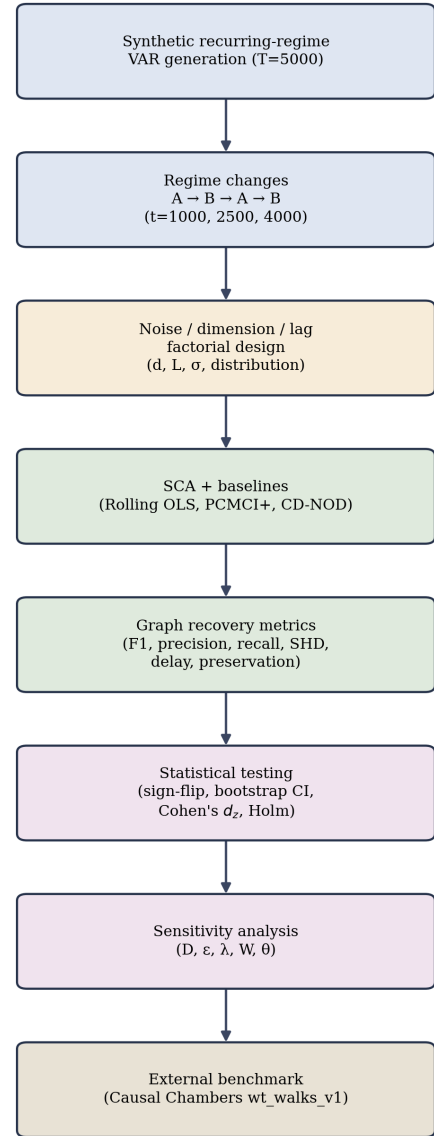


FIGURE 4. Experimental pipeline from synthetic-data generation through the external benchmark.

this canonical condition, CD-NOD's regime-aware summary-graph approach recovers more of the true edge set than SCA does, at the cost of following the graph less conservatively.

CD-NOD's adaptation delay of 500.0 (the length of the post-change evaluation window) indicates that its change-point procedure did not resolve a discrete adaptation time within the evaluated horizon for this cell, which should temper direct comparison of the delay column.

B. POOLED FACTORIAL RESULTS

Table 2 shows representative cells drawn from the full factorial design (held-out seeds, mean $\pm$ s.d. where available), and Table 3 reports the paired statistical comparison between SCA and Rolling OLS pooled across all factorial conditions and held-out seeds.

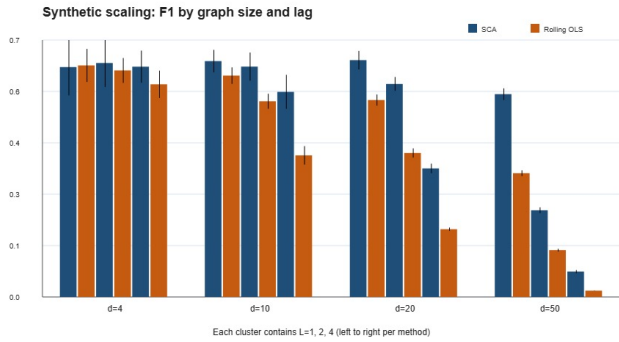
Stable-edge preservation for these same cells (not shown

TABLE 1. Canonical Dynamic Comparison: SCA vs. Rolling OLS vs. CD-NOD ($d = 4, L = 2$, Gaussian, $\sigma = 0.10$; mean $\pm$ s.d.)

Method	Prec.	Recall	F1	SHD	Delay	Stable
SCA	0.869 $\pm$ 0.042	0.551 $\pm$ 0.081	0.637 $\pm$ 0.065	2.508 $\pm$ 0.270	456.3 $\pm$ 51.5	0.626 $\pm$ 0.116
Rolling OLS	0.806 $\pm$ 0.037	0.537 $\pm$ 0.042	0.617 $\pm$ 0.034	2.352 $\pm$ 0.154	386.5 $\pm$ 81.9	0.582 $\pm$ 0.097
CD-NOD	0.638 $\pm$ 0.093	1.000 $\pm$ 0.000	0.775 $\pm$ 0.069	2.400 $\pm$ 0.940	500.0 $\pm$ 0.0	1.000 $\pm$ 0.000

TABLE 2. Representative Factorial Performance (held-out seeds)

Condition	Method	Prec.	Rec.	F1	SHD	Delay
$d=4, L=1, \sigma=0.05$	Rolling	0.934	0.891	0.906	0.640	173.7
$d=4, L=1, \sigma=0.05$	SCA	0.917	0.908	0.908	0.697	305.3
$d=4, L=1, \sigma=0.10$	Rolling	0.873	0.526	0.631	2.090	401.8
$d=4, L=1, \sigma=0.10$	SCA	0.878	0.522	0.626	2.152	483.3
$d=4, L=1, \sigma=0.25$	Rolling	0.165	0.049	0.074	3.975	500.0
$d=4, L=1, \sigma=0.25$	SCA	0.041	0.029	0.028	4.075	500.0
$d=50, L=1, \sigma=0.05$	SCA	0.737	0.921	0.817	20.358	—
$d=50, L=2, \sigma=0.05$	SCA	0.246	0.883	0.384	142.407	—
$d=50, L=4, \sigma=0.05$	SCA	0.067	0.751	0.121	509.517	—

**FIGURE 5. Synthetic scaling: mean F1 by graph dimension (d) and lag order ($L = 1, 2, 4$ within each cluster) for SCA vs. Rolling OLS, illustrating the F1 collapse at high d and L described in Section VI-B.**

in Table 2 for space) is 0.939/0.993 (Rolling/SCA) at $d = 4$, $\sigma = 0.05$; 0.575/0.587 at $d = 4$, $\sigma = 0.10$; 0.067/0.011 at $d = 4$, $\sigma = 0.25$; and 0.968, 0.928, 0.847 for SCA at $d = 50$ with $L = 1, 2, 4$ respectively.

Two trends emerge in Table 2: first, as dimension and noise drop to low values ($d = 4$, $\sigma = 0.05$), SCA and Rolling OLS achieve similar F1, though SCA exhibits a considerably longer adaptation delay (305.3 vs. 173.7 steps), due to the conservative bias of the persistence gate: SCA waits for a substantial signal before declaring a change, whereas Rolling OLS always reacts to a new data window; second, with increasing dimension ($d = 50$), there is a severe drop in F1 score as the lag order L rises from 1 to 4 (0.817 $\rightarrow$ 0.384 $\rightarrow$ 0.121), mostly attributable to a large decrease in precision (0.737 $\rightarrow$ 0.246 $\rightarrow$ 0.067), while recall remains relatively high; correspondingly, SHD increases by more than an order of magnitude (20.4 $\rightarrow$ 142.4 $\rightarrow$ 509.5), reflecting the higher number of candidate edges at large d and L . Section VI-G demonstrates that the $L = 4$ cell also corresponds to a failure in change detection.

TABLE 3. Pooled SCA vs. Rolling OLS Statistical Comparison (all factorial cells, held-out seeds)

Metric	Δ	95% CI	d_z	Holm p
Precision	+0.0774	[0.0762, 0.0787]	26.21	0.00012
Recall	-0.0206	[-0.0212, -0.0199]	-14.20	0.00012
F1	+0.0498	[0.0492, 0.0505]	34.77	0.00012
SHD	-244.20	[-244.44, -243.95]	-419.36	0.00015
Delay	+0.575	[-0.507, 1.757]	0.22	0.71227
Stable	-0.0106	[-0.0119, -0.0091]	-3.19	0.00012

Pooled across the full factorial design, SCA shows a statistically significant improvement over Rolling OLS in precision, F1, and SHD (all Holm $p < 0.001$), at the cost of a small but significant reduction in recall and in stable-edge preservation (both Holm $p < 0.001$). The pooled SHD difference of -244.20 is large in absolute terms, but this magnitude is driven substantially by the high- d , high- L cells in which SHD scales with the number of candidate edges ($d^2(L+1)$); it should not be read as a uniform per-cell improvement of that size. The aggregate adaptation-delay difference (+0.575 steps) is not statistically distinguishable from zero (Holm $p = 0.712$, $d_z = 0.22$): averaged across the whole factorial design, SCA is neither reliably faster nor reliably slower than Rolling OLS at fully recovering a post-change edge set, even though the two methods diverge substantially in individual cells (Table 2).

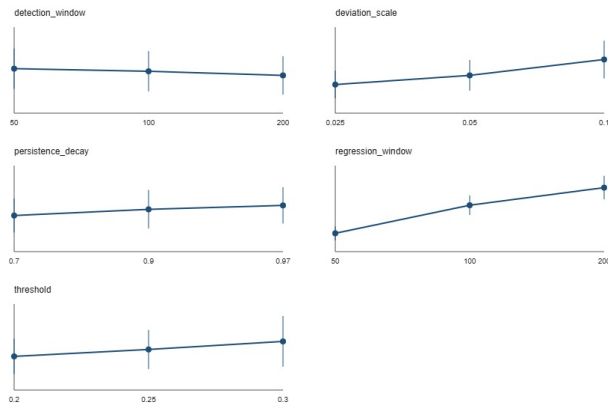
C. SENSITIVITY ANALYSIS

Table 4 reports a one-factor-at-a-time sensitivity sweep over SCA's five principal hyperparameters, performed on the tuning seeds (0–19) and held fixed at their selected value elsewhere in the row.

The practical tradeoffs are similar for different parameters. Smaller change-detection window D leads to slightly better F1 but worse SHD, since a shorter window has more noise. Increasing edge-change tolerance ϵ from 0.025 to 0.10 monotonically improves both F1 and SHD, since it suppresses false positives due to estimation noise; the same reasoning applies to raising the detection threshold θ , which also improves F1 while drastically reducing the false-alarm rate (0.089 $\rightarrow$ 0.017). The persistence decay λ has less impact, with higher λ (less decay, i.e., longer memory of past deviations) giving slightly better F1. The biggest impact is from the local-estimation window W : doubling it from 50 to 100 and again to 200 improves F1 by a significant margin (0.408 $\rightarrow$ 0.531 $\rightarrow$ 0.609) and roughly halves the SHD, at the cost of slower response to a real change (the usual bias–

TABLE 4. Sensitivity Analysis (tuning seeds; one factor varied at a time)

Parameter	Value	F1	SHD	FAR
D (window)	50	0.537	3.604	—
D (window)	100	0.531	3.854	—
D (window)	200	0.523	4.363	—
ϵ (tolerance)	0.025	0.506	4.261	—
ϵ (tolerance)	0.05	0.531	3.854	—
ϵ (tolerance)	0.10	0.575	3.225	—
λ (decay)	0.70	0.517	4.077	—
λ (decay)	0.90	0.531	3.854	—
λ (decay)	0.97	0.540	3.762	—
W (local win.)	50	0.408	6.051	—
W (local win.)	100	0.531	3.854	—
W (local win.)	200	0.609	2.824	—
θ (threshold)	0.20	0.531	3.858	0.089
θ (threshold)	0.25	0.547	3.734	0.057
θ (threshold)	0.30	0.567	3.608	0.017

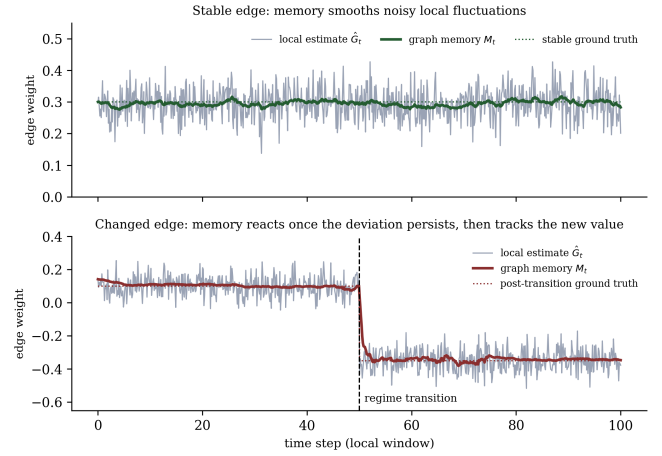
Hyperparameter sensitivity (mean F1 $\pm$ SD; tuning seeds 0–19)**FIGURE 6. Hyperparameter sensitivity: mean F1 $\pm$ SD across tuning seeds (0–19) for each of SCA’s five principal hyperparameters (Section VI-C).**

variance tradeoff for a sliding-window estimator). Note that these results only reflect the single-variable tradeoffs, since we varied one parameter at a time; Section VIII discusses interactions between parameters.

D. ABLATION STUDY

The development benchmark (regime_change_a1.csv: $T = 5000$, $d = 4$, $L = 2$, Gaussian noise scale 0.1, seed 42, single change at $t = 2500$) is a smaller, earlier diagnostic instance where three lagged/stable relationships are unchanged across the transition, one contemporaneous edge is removed, and one lagged edge is added. Since the estimator is past-only and cannot recover contemporaneous directions, this benchmark is treated as a diagnostic smoke test instead of primary manuscript evidence, and Table 5 reports the before/after metrics for the full SCA model and two ablations.

The unnecessary-change rate follows the same ordering: 0.0409 (Full SCA), 0.0653 (No Persistence), 0.0813 (Global Only), and 0.2397 (Rolling regression); PCMCi+ is static

**FIGURE 7. Conceptual edge trajectory: local estimate vs. graph memory. Memory smooths a stable edge’s noisy local fluctuations (top) but, once a deviation is detected and persists, tracks a changed edge to its new value (bottom).**

and has no post-change or unnecessary-change entry. Removing the persistence gate (“No Persistence”) and then removing edge-level selectivity entirely (“Global Only”, a single graph-wide plasticity rate) both degrade post-change F1 and stable-edge preservation monotonically relative to the full model, confirming that both mechanisms contribute to the observed gains on this benchmark. Rolling regression recovers the pre-change local graph essentially perfectly (F1=1.0000, SHD=0), which is unsurprising since it has no historical commitment to overturn; however, its stable-edge preservation is substantially worse than SCA’s (0.7603 vs. 0.9591) and its rate of unnecessary changes is roughly six times higher (0.2397 vs. 0.0409), while its adaptation delay of 352 steps is over four times SCA’s 80 steps on this particular instance. We do not read this single, small development instance as general evidence that SCA dominates Rolling OLS—the pooled factorial comparison in Table 3 shows a more nuanced picture—but rather as confirmation that the ablated mechanisms behave as designed: SCA is more conservative and, on this instance, correspondingly slower to fully match the pre-change graph, while offering better protection of unchanged structure and fewer unnecessary changes than an un-gated update.

E. PLASTICITY ANALYSIS

Table 6 reports the average realised plasticity rate on changed edges versus stable edges for the full model and the two ablations from Section VI-D.

The Global-Only ablation applies the same rate to changed and stable edges by construction (ratio 1.000 $\times$), confirming that it has no edge-level selectivity. Adding persistence decreases both the changed- and stable-edge rates substantially compared to No Persistence (0.0349 $\rightarrow$ 0.0231 and 0.0307 $\rightarrow$ 0.0204, respectively), but only slightly changes the ratio between them (1.137 $\times$ $\rightarrow$ 1.132 $\times$). This is a genuinely nuanced finding: persistence’s main empirical effect on this

TABLE 5. Development Benchmark / Ablation (regime_change_a1.csv)

Method	F1 pre	F1 post	SHD pre	SHD post	Stable	Detect.	Delay
Full SCA	0.5714	0.7273	6	3	0.9591	0.8712	80
No Persist.	0.4615	0.6667	7	4	0.9347	0.7780	41
Global Only	0.4286	0.5333	8	7	0.9187	0.7580	27
Rolling regr.	1.0000	0.7500	0	2	0.7603	0.8592	352
PCMCI+ (static)	0.8889	N/A	1	N/A	—	—	—

TABLE 6. Plasticity Analysis: Changed- vs. Stable-Edge Update Rates

Method	Changed	Stable	Ratio
Full SCA	0.0231	0.0204	1.132×
No Persistence	0.0349	0.0307	1.137×
Global Only	0.1400	0.1400	1.000×

TABLE 7. Causal Chambers External Benchmark (actuators_random_walk_2)

Method	TP/FP/FN	Prec.	Rec.	F1	SHD	Time (s)
SCA	24/125/18	0.161	0.571	0.251	143	56.0
PCMCI+	1/33/41	0.029	0.024	0.026	74	87.1

benchmark is to scale down plasticity overall—to act as a conservative brake on both changed and stable edges alike—rather than to sharply raise the amount by which changed edges are treated differently than stable ones. Edge-level selectivity (comparing Global Only to No Persistence) is what causes most of the ratio’s departure from 1.0× in the first place.

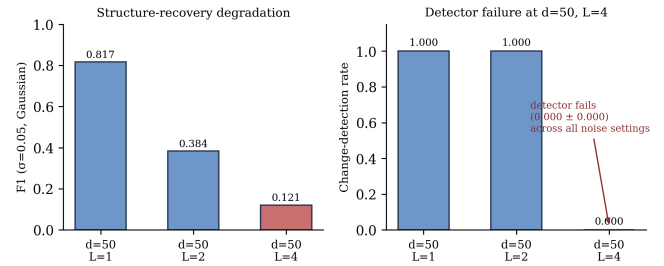
F. EXTERNAL BENCHMARK

Table 7 reports results on the wt_walks_v1/actuators_random_walk_2 Causal Chambers series (32 variables, 10,000 observations, lags 1–10 collapsed to lag-agnostic directed-edge recovery).

SCA achieves a much higher F1 score than PCMCI+ on this dataset (0.251 compared to 0.026), mainly due to its higher recall value; however, PCMCI+’s SHD is also lower (74 versus 143) because its predictions are much sparser than SCA’s. In fact, fewer predicted edges result in fewer false positives, but such an inference implies that most of the true edges are missed by PCMCI+. We do not interpret PCMCI+’s lower SHD as a sign of being a better algorithm in this case, nor do we make generalisations based on a single data set that may not justify any type of conclusion about real-life efficacy of any of these methods.

G. FAILURE MODES

The change detector has a detection rate of 1.000 ± 0.000 in all factorial cells except for $d = 50$, $L = 4$, where it collapses to 0.000 ± 0.000 across all evaluated noise levels and noise distributions. This coincides with the drastic degradation in F1 and SHD seen for $d = 50$, $L = 4$ in Table 2, where F1=0.121 and SHD=509.5. This can be explained by the top-

**FIGURE 8.** SCA structure-recovery degradation and change-detector failure as dimension and lag order increase at $d = 50$ ($\sigma = 0.05$, Gaussian).

k design where $k = \min(4, d^2(L + 1))$, and thus at $d = 50$ and $L = 4$ the number of candidate edges is so high that four slots will not be able to capture the truly changed entries above the noise level of many little unchanged candidate coefficients. We consider this an important limitation rather than an incident, and it is further discussed in Section VIII.

The false-alarm rates across the factorial design that are not discussed in this section fall between approximately 0.091 and 0.097 for the canonical and the representative conditions summarised in Table 1. A separate no-change control experiment ($T = 2000$, $d = 4$, $L = 2$, $\sigma = 0.1$, no true regime changes, 20 seeds) yielded a significantly lower false-alarm rate of 0.0143 ± 0.0047 (about 1.4%). However, these two false-alarm rates are not directly comparable: the ~9% rate is computed on a series that has undergone regime changes (which might include false alarms just before or after the proper change), while the 1.4% rate comes from a stationary control series with no changes to detect.

VII. DISCUSSION

Selective, edge-level adaptation using memory and persistence is the contribution of SCA. Rather than claiming overall superiority over all competing methods, the data presented here demonstrate that (i) SCA improves precision and F1 over Rolling OLS with only a minor loss in recall, as shown in Table 3; (ii) somewhat disappointingly, SCA offers no improvement over Rolling OLS in stable-edge preservation, actually showing an average difference of -0.0106 (Holm $p = 0.00012$), which suggests that the memory mechanism used by SCA is not better, in terms of stable-edge preservation, than simple re-estimation at every step, despite the demonstrable benefit over Rolling OLS on the small development benchmark of Table 5; and (iii) in the comparison of overall

adaptation delay, SCA and Rolling OLS are approximately on the same level, with neither method significantly better than the other across the factorial design ($p = 0.71227$): throughout the factorial design, neither algorithm was reliably faster in its ability to fully adapt to a new edge set following a change point, despite noticeable differences in delay in individual cells and in the development benchmark.

In the canonical dynamic scenario, the F1 performance of CD-NOD exceeds that of SCA (0.775 vs. 0.637) because CD-NOD has perfect recall in that cell. Meanwhile, SCA trades lower recall for higher precision and a shorter adaptation delay than CD-NOD. Rolling OLS does very well in terms of the raw score of local-graph recovery. In particular, it performs perfectly before the occurrence of a change (see Table 5, $F1=1.0000$ for the development benchmark), which is not surprising because it uses only the most recent information and does not retain any historical information that contradicts it. SCA uses a different approach, which gives it better resistance to unnecessary changes (Table 5, unnecessary-change rate of 0.0409 as opposed to 0.2397 for Rolling OLS), but introduces inertia that leads to longer adaptation delays in some cells (see Table 2), even though in the aggregate comparison those delay differences are not significant.

High-dimensional, high-lag settings expose an important failure mode that must be discussed explicitly rather than implicitly: at $d = 50, L = 4$ the change detector's detection rate collapses to 0.000 across every evaluated noise condition (Section VI-G), coinciding with severe F1 and SHD degradation. This traces to the detector's fixed top- k design, which becomes under-powered when the number of candidate edges ($d^2(L + 1)$) grows large compared to the fixed number of slots examined. The external Causal Chambers benchmark (Section VI-F) contains only one physical series, so its results should be read as an illustrative check rather than a validated real-world evaluation; and the current experiments do not fully test nonlinear generating mechanisms or heteroscedastic noise, both of which are addressed by generative frameworks such as FANTOM [8] but are outside the present study's scope. The one-factor-at-a-time sensitivity analysis (Section VI-C) characterises each hyperparameter's marginal effect, but not their interactions. Finally, unlike Rhino [4] or FANTOM [8], the present work provides no theoretical convergence, bias, or identifiability proof for the persistence-gated memory filter; SCA's justification is empirical, and its use of a regression-based evidence score as a proxy for edge presence relies informally on established identifiability results for additive-noise causal models [8], [14] rather than on a dedicated proof of its own.

VIII. LIMITATIONS AND FUTURE WORK

A. LIMITATIONS

We summarise the limitations discussed throughout the paper in one place. (1) SCA is a past-only, Granger-style estimator: the lag-zero plane is fixed to zero and makes no claim about contemporaneous structural causal identification. (2) The change detector fails outright in the high-dimensional,

high-lag regime tested here ($d = 50, L = 4$), where its detection rate drops to zero. (3) The detector's fixed top- k design does not scale its sensitivity to the number of candidate edges, which is the root cause of (2). (4) Nonlinear generating mechanisms are not evaluated; all synthetic data are generated by a linear VAR process. (5) Heteroscedastic noise, of the kind explicitly modelled by FANTOM [8], is not evaluated. (6) The external validation rests on a single Causal Chambers physical series, too little data to make a general real-world reliability claim. (7) RPCMCI, CASTOR, and FANTOM are discussed as related methods but were not run end-to-end on this study's factorial design, so no quantitative comparison against them is reported here. (8) The sensitivity analysis varies one hyperparameter at a time and cannot characterise interactions between parameters such as D, ϵ, λ, W , and θ . (9) No theoretical convergence, bias, or identifiability guarantee is provided for the persistence-gated memory filter.

B. FUTURE WORK

Several directions follow from the above limitations: (1) an adaptive or dimension-aware change detector, for example one that scales its score by the number of candidate edges or calibrates k to d and L rather than using a fixed small constant; (2) direct factorial sweeps over selectivity and plasticity bounds ($[\alpha_{\min}, \alpha_{\max}]$) rather than the current one-factor-at-a-time protocol to expose interaction effects; (3) evaluation on nonlinear and heteroscedastic data generators, potentially by replacing the OLS-based local evidence score with a learned, history-dependent likelihood in the spirit of Rhino [4] or FANTOM's conditional normalising flows [8] at the cost of the current pipeline's training-free simplicity; (4) additional real-world benchmarks beyond the single Causal Chambers series used here, including EEG or other biomedical series with documented heteroscedastic noise [8], [13]; (5) a full quantitative comparison against RPCMCI, CASTOR, and FANTOM on a shared factorial design; (6) multi-factor hyperparameter analysis (e.g., a designed experiment or Bayesian optimisation over the joint hyperparameter space); (7) theoretical analysis of the convergence and bias properties of the persistence-gated exponential memory filter; and (8) regime-aware memory mechanisms that combine SCA's edge-level selectivity with an explicit, discrete notion of regime identity, potentially recovering some of CD-NOD's and FANTOM's recall advantage without discarding SCA's precision and conservatism.

IX. CONCLUSION

This paper introduces Selective Causal Adaptation, a training-free streaming framework which shifts the unit of adaptation in dynamic causal discovery from the graph as a whole to the individual edge, gating a persistent causal-graph memory with a multiplicative, edge-level plasticity rate that combines the impact of global change intensity, local deviation, and edge-specific persistence. Under the evaluated conditions, while SCA consistently improves precision and pooled F1 compared to a memory-free Rolling-OLS baseline at the cost

of recall, it does not demonstrate a universal advantage in either stable-edge preservation or adaptation delay once results are aggregated across the full factorial design; CD-NOD achieves a higher F1 than SCA on the canonical dynamic condition by virtue of a significantly better recall; and a high-dimensional, high-lag stress test reveals a change-detector failure mode on which we advise caution as an honestly reported limitation rather than an exceptional capability. Two ablation experiments confirm the contribution of both the persistence gate and edge-level selectivity to the observed performance, while a detailed plasticity analysis reveals that persistence manifests primarily as a conservative dampening of adaptation rate, rather than a dramatic increase in contrast between changed and stable edges. We posit SCA as a lightweight alternative in applications where non-stationarity is expected to manifest at the edge level and a training-free update regime is desirable, rather than a replacement for regime-aware generative models such as CASTOR or FAN-TOM in settings which benefit from their greater flexibility and theoretical guarantees.

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REFERENCES

- [1] J. Runge, "Discovering contemporaneous and lagged causal relations in autocorrelated nonlinear time series datasets," in *Proc. Conf. Uncertainty Artif. Intell. (UAI)*, PMLR, 2020, pp. 1388–1397.
- [2] X. Zheng, B. Aragam, P. K. Ravikumar, and E. P. Xing, "DAGs with NO TEARS: Continuous optimization for structure learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 31, 2018.
- [3] R. Pamfil, N. Sriwattanaworachai, S. Desai, P. Pilgerstorfer, K. Georgatzis, P. Beaumont, and B. Aragam, "DYNOTEARS: Structure learning from time-series data," in *Proc. Int. Conf. Artif. Intell. Statist. (AISTATS)*, PMLR, 2020.
- [4] W. Gong, J. Jennings, C. Zhang, and N. Pawłowski, "Rhino: Deep causal temporal relationship learning with history-dependent noise," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.
- [5] B. Huang, K. Zhang, J. Zhang, J. Ramsey, R. Sanchez-Romero, C. Glymour, and B. Schölkopf, "Causal discovery from heterogeneous/nonstationary data," *J. Mach. Learn. Res.*, vol. 21, no. 89, pp. 1–53, 2020.
- [6] E. Saggioro, J. de Wiljes, M. Kretschmer, and J. Runge, "Reconstructing regime-dependent causal relationships from observational time series," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 30, no. 11, 2020.
- [7] A. Rahmani and P. Frossard, "Causal temporal regime structure learning," in *Proc. 28th Int. Conf. Artif. Intell. Statist. (AISTATS)*, PMLR, vol. 258, 2025, pp. 4546–4554.
- [8] A. Rahmani and P. Frossard, "Flow based approach for dynamic temporal causal models with non-Gaussian or heteroscedastic noises," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025.
- [9] S. Mameche, L. Cornanguer, U. Ninad, and J. Vreeken, "SPACETIME: Causal discovery from non-stationary time series," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 18, 2025, pp. 19405–19413.
- [10] D. Xu, C. Ruan, E. Korpeoglu, S. Kumar, and K. Achan, "Inductive representation learning on temporal graphs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.
- [11] M. H. Ferdous, E. Hossain, and M. O. Gani, "TimeGraph: Synthetic benchmark datasets for robust time-series causal discovery," in *Proc. 31st ACM SIGKDD Conf. Knowl. Discovery Data Mining (KDD)*, 2025.
- [12] J. L. Gamella, J. Peters, and P. Bühlmann, "Causal chambers as a real-world physical testbed for AI methodology," *Nature Mach. Intell.*, vol. 7, no. 1, pp. 107–118, 2025.
- [13] H. M. Huizenga and P. C. Molenaar, "Equivalent source estimation of scalp potential fields contaminated by heteroscedastic and correlated noise," *Brain Topogr.*, vol. 8, pp. 13–33, 1995.
- [14] J. Peters, J. M. Mooij, D. Janzing, and B. Schölkopf, "Causal discovery with continuous additive noise models," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 2009–2053, 2014.
- [15] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

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